

Laplace Transform Formulas

$f(t)$	$F(s)$
1	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(at)$	$\frac{a}{s^2 + a^2}$
$\cos(at)$	$\frac{s}{s^2 + a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\mathcal{U}(t-a)$	$\frac{e^{-as}}{s}$
$\mathcal{U}(t-a)f(t-a)$	$e^{-as}F(s)$
$e^{at}f(t)$	$F(s-a)$
$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n} F(s)$
$\sin(at) - at \cos(at)$	$\frac{2a^3}{(s^2 + a^2)^2}$
$t \sin(at)$	$\frac{2as}{(s^2 + a^2)^2}$
$f * g$	$F(s)G(s)$
$\delta(t-a)$	e^{-sa}
$f'(t)$	$sF(s) - f(0)$
$f''(t)$	$s^2F(s) - sf(0) - f'(0)$

Cover-Up/Limit Method

- Distinct Linear Factors:

$$\frac{P(x)}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

To find the coefficients, we use:

$$A = \lim_{x \rightarrow a} \left[(x-a) \frac{P(x)}{(x-a)(x-b)(x-c)} \right]$$

- “Cover up” factor in denominator
- Evaluate at the “root”

- **Ex:**

$$\frac{5x+3}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$$

$$A = \left. \frac{5x+3}{x-2} \right|_{x=1} = -8$$

$$B = \left. \frac{5x+3}{x-1} \right|_{x=2} = 13$$

- Repeated Linear Factors (Derivative Limit Method):

$$F(x) = \frac{P(x)}{G(x)(x-a)^n} = \frac{A_1}{x-a} + \underbrace{\frac{A_2}{(x-a)^2} + \dots + \frac{A_n}{(x-a)^n}}_{\text{factors of } (x-a)^n} + \dots \text{ (for the } G(x) \text{ term(s))}$$

For the k^{th} coefficient:

$$A_k = \frac{1}{(n-k)!} \lim_{x \rightarrow a} \left\{ \frac{d^{n-k}}{dx^{n-k}} [(x-a)^n F(x)] \right\}$$

- When $k = n$ (the highest order factor), we have the same formula as for distinct linear factors.
- We use the derivative to “recover” the lower order roots.

- **Ex:**

$$F(x) = \frac{2x+3}{(x-1)^3(x+2)} = \frac{A_1}{(x-1)} + \frac{A_2}{(x-1)^2} + \frac{A_3}{(x-1)^3} + \frac{B}{x+2}$$

$$A_3 = \left. \frac{2x+3}{x+2} \right|_{x=1} = \frac{5}{3} \qquad A_1 = \left. \frac{1}{2!} \frac{d^2}{dx^2} \left(\frac{2x+3}{x+2} \right) \right|_{x=1} = -\frac{1}{27}$$

$$A_2 = \left. \frac{d}{dx} \left(\frac{2x+3}{x+2} \right) \right|_{x=1} = \frac{1}{9} \qquad B = \left. \frac{2x+3}{(x-1)^3} \right|_{x=-2} = -\frac{1}{27}$$

Note:

- For irreducible quadratic factors, this doesn’t apply (use $Cx + D$ and solve).
- Always check that (deg of num) < (deg of denom).